

令和 9(2027)年度
東京大学大学院学際情報学府
学際情報学専攻

表現

修士課程（夏季募集）

研究計画書

受験番号 ●— この欄は記入しないこと

氏名

Shulamith Rivera Sevilla

研究テーマ ●— 希望する研究テーマを 40 字以内（英語 25 words 程度）で簡潔に記入すること。

SoftVoicesAI: Thoughtful Agents to Inspire Critical Reflection on Social Media Takes

注：入学願書と同じ内容とすること。

キーワード ●— 5 単語程度

Human-AI Interaction; Large Language Models; Social Media; Critical Thinking; Integrative Complexity

研究要旨 ●— 200 字程度（英語 120 words 程度）

"AI is making us dumber." "Kids today can't focus." Takes like these dominate social media, where feeds reward outrage and pushing back makes people defensive, not reflective. This proposal presents SoftVoicesAI, an interface where multiple AI agents comment on both a social media post and the user's response to it. Each agent plays a cognitive role, examining the take from a different angle — such as a hidden assumption, a missing perspective, or a leap in reasoning — and speaks in a communication style modeled on high-quality listening: for instance, acknowledging what's valid before adding nuance, or offering considerations rather than conclusions. The user remains the judge of their own view. The study tests whether reading the agents' comments is associated with increased integrative complexity — a marker of critical reflection — and examines how users experience the interface.

次ページ以下に任意様式で 4 頁以内に研究計画書をまとめて記載すること。

SoftVoicesAI: Thoughtful Agents to Inspire Critical Reflection on Social Media Takes

1. Problem and Objective

"AI is making us dumber." "Kids today can't focus." "Work from home is killing your career." Takes like these are everywhere on social media. In an audit of X/Twitter's feed, the engagement-based ranking algorithm showed people more emotionally charged, hostile content than a simple newest-first feed would have — content that users themselves said made them feel worse about the other side [1]. Analyses of over 100 million posts show that on platforms like X/Twitter, people largely interact inside echo chambers: clusters of like-minded users where existing views get repeated back rather than examined [2].

The difficulty is not only the environment but also how minds work. People typically form judgments quickly and intuitively, then produce reasons afterward to justify what they already believe [3]. On top of that, pushing directly against someone's view backfires: being told "you're wrong, and here's why" triggers anger and counterarguing that makes people reject the message outright [4]. In a field experiment, X/Twitter users paid to follow accounts from the opposing political side for a month did not moderate their views — if anything, they became more extreme [5].

Encouraging critical reflection is therefore two problems at once:

- **A content problem** — what is worth surfacing about a given take? Which assumptions, missing perspectives, or weak links in reasoning deserve attention?
- **A delivery problem** — how can those things be surfaced so that the reader stays open and curious, instead of getting defensive and shutting the conversation down?

The objective of this research is to design and empirically evaluate SoftVoicesAI, an AI-assisted tool that addresses both problems together, designed to promote critical reflection on takes found on social media.

2. Related Work

Human–computer interaction research increasingly explores how AI might support thinking rather than replace it. SoftVoicesAI builds on several lines of research:

AI that prompts thinking instead of answering. *Yatani et al.* propose Extraheric AI: AI that poses questions and alternative perspectives so the user exercises their own higher-order thinking skills [6]. *Danry et al.* showed that people spot flawed logic more accurately when an AI frames its input as questions to consider, rather than as explanations telling them the answer [7].

Multiple AI agents with different voices. *Fang et al.* built an interface where AI agents exchange thoughts with junior researchers during paper reading; the exchanges improved critical thinking scores compared with reading without agents, and multiple agents drew more perspective-comparison than one [8]. *Liu et al.*'s Perspectra let users convene simulated domain experts in a forum-style discussion, producing deeper critical engagement than a plain group chat [9].

Agents that challenge the user. *Tanprasert et al.* deployed chatbots that debate users about YouTube videos, and found that the bot's speaking style and social identity changed how much critical thinking the debate produced [10]. *Chiang et al.* showed that an AI playing devil's advocate can help groups rely more appropriately on AI recommendations [11].

AI against misinformation and false belief. *Szabó et al.* showed that a chatbot conversation can "inoculate" people against misinformation more effectively than traditional reading or writing exercises [12]. *Costello et al.* found that short, personalized dialogues with an AI reduced belief in conspiracy theories, with the effect persisting at follow-up [13]. That line of work aims at persuasion: moving people toward a predetermined correct belief. SoftVoicesAI aims at something different — reflection without a target: its agents offer considerations, not conclusions, and the user remains the judge of their own view.

Interface design alone can also help: even before LLMs, *Kriplean et al.*'s ConsiderIt showed that organizing civic debate around pro/con points that users could adopt led voters to write positions acknowledging both sides [14].

SoftVoicesAI brings these together: multiple AI agents respond to takes on social media, each examining the claim from a distinct angle (its cognitive role) and speaking in a way designed to invite rather than persuade (its communication style).

3. The Proposed System: SoftVoicesAI

SoftVoicesAI is an interface where multiple AI agents comment on both a social media post and the user's response to it. Each agent examines the take from a different angle and speaks in a style modeled on high-quality listening (details in 3B and 3C). An interactive mockup is available at <https://mithi.github.io/mockups/softvoices-ai>.

A. The SoftVoicesAI Pipeline. Participants respond to a real X/Twitter-style post, read AI-generated comments, then revisit their initial response. Each session follows five steps:

1. **Stimulus.** A real post known to have drawn attention and polarized response, with brief context (e.g., surrounding events) where useful; post selection is detailed in Section 5C.
2. **Prompt.** A fixed open-ended question: *"What is this post claiming or implying? What's your view on that and why? Feel free to share any other thoughts."*
3. **Initial response.** The user writes their take (≥ 300 characters; this threshold will be validated for reliable integrative complexity (IC) scoring in the pilot).
4. **Agent comments.** Agents whose cognitive role applies to the post (Section 3C) comment on both the post and the user's response, two at a time in a fixed order, expanded via a *Load more* button. Users may tag each comment with private reactions or flag it (Section 3D).
5. **Revisit.** Once all comments are shown, the user writes a final, standalone response to the original question — framed to make clear their view may stay, shift, or evolve.

The session ends there; no follow-ups are offered.

B. Communication Style. How the agents speak follows a specification grounded in research on high-quality listening [19]. For instance, agents acknowledge what's valid before adding nuance, offer considerations rather than verdicts, and prefer tentative phrasing over forceful claims, which tend to put people on the defensive [4]. Take a viral post claiming an MIT brain-scan study proves ChatGPT is making users "cognitively bankrupt." A forceful reply might say: *"Boomers said this about TV. Every generation invents a new thing to blame for being stupid."* An agent says instead: *"The worry makes sense — the numbers do sound alarming. Though similar things were said about calculators and Google — that kind of pattern might be worth looking into. That said, AI creates things, so maybe this one really is different. What do you think?"* The communication style guideline is at <https://mithi.github.io/articles/how-to-invite-reflection>.

C. Cognitive Roles. Each agent plays a cognitive role, examining the take from a different angle — such as a hidden assumption, a missing perspective, or a leap in reasoning. For the same post in Section 3B, an agent with the Warrant Surfacers role might comment: *"The finding is striking — 'less brain activity' is a real result. Though the jump from that to 'we're getting dumb' is a big one; less activity could also mean the brain is working smarter — offloading, not declining. Thoughts?"* The role specifications and example responses — combining ideas from cognitive psychology, argumentation theory, media studies, and meta-science [15–18] — are at <https://mithi.github.io/articles/cognitive-roles>. Roles appear in a fixed order that opens with acknowledgment, then broadens the take, then examines it more precisely — an application of the high-quality listening principle [19] at the session level. Not every role applies to every post; in each session, an LLM judges which roles fit the post and the user's response.

D. Tagging Agent Comments with Reactions and Flagging. Users may tag each comment with a private reaction —  raised something new,  worth sitting with,  shifted my view,  didn't land for me — or flag it as factually questionable (). The reactions look like familiar social media reaction buttons, but the counts stay private, so no comment becomes a "winning side" to rally around. Flagged comments remain visible and are recorded for analysis.

E. Comment Generation and Quality Control. Comments are generated by an LLM, conditioned on the post, its context, and the user's initial response, and kept short to match the feel of social media replies. LLMs can hallucinate; rather than reviewing each comment before it is shown — which would mask whether users notice inaccuracies on their own — participants are told during onboarding to treat the agents' factual claims with the same skepticism as any social media comment, and the pilot screens for systematic hallucination patterns before the main study begins.

4. Research Questions

RQ1. Is reading the agents' comments associated with an increase in integrative complexity (IC, a measure of critical reflection) from initial to revisited responses? **RQ2.** How do users experience the interface in terms of cognitive load — whether mental effort feels productive (germane) or wasted (extraneous) — perceived effectiveness, enjoyability, and willingness to adopt? **RQ3.** What patterns emerge across sessions, agent roles, response length, reaction and flagging behavior, and interview themes?

5. Study Design

A. Pilot Study. A pilot with 3–5 participants will refine the prompts and stimuli, train raters to inter-rater reliability, check that responses of ≥ 300 characters support reliable IC scoring, screen agent comments for systematic hallucination patterns, estimate session duration, and a power analysis to confirm or increase the planned sample.

B. Participants. About 24 adults, all active social media users (daily or near-daily) who self-report wanting to improve their critical thinking. No demographic targeting. Around 32 will be recruited to allow for attrition; participants missing more than one session are excluded from the primary analysis. All participants are compensated for their time.

C. Stimuli and Procedure. Six real posts serve as stimuli, curated by the researchers from public engagement signals (e.g., reply volume, cross-platform spread) and chosen to vary in topic and type of claim; real posts are used so the task mirrors what people actually encounter. Sensitive topics such as race, gender, and sexual orientation are excluded. The study is within-subjects across eight sessions.

- **Session 0** (~15 min) covers consent, a questionnaire on demographics and social media habits, the note about agent factual errors (Section 3E), and one practice session that is not analyzed.
- **Sessions 1–6** (~15 min each, on separate days) each present one post through the full pipeline, followed by a questionnaire. Post order is rotated across participants using a balanced 6×6 Latin square (6 posts × 6 positions, 4 participants per row) so effects of order don't get tangled with effects of specific posts.
- **Session 7** (~20 min) is an end-of-study questionnaire and a semi-structured interview.

D. Measures. For **RQ1**, initial and revisited responses are scored for IC (1–7) using the *Baker-Brown et al.* manual [20], by two raters blind to participant identity and to whether a response is initial or revisited (weighted κ reported). Full blinding cannot be guaranteed — revisited responses may be longer or mention points the agents raised — so this is monitored during pilot rater training. For **RQ2**, each session ends with six single-item ratings, kept minimal to limit burden across six sessions: overall mental effort (the Paas scale), two cognitive-load items — how much effort felt productive (germane) and how much felt wasted (extraneous) — and one Likert item each for perceived effectiveness, enjoyability, and willingness to adopt. The end-of-study questionnaire uses multi-item versions of these, plus role-level perceived usefulness and items on whether engagement habits changed outside the study. For **RQ3**, reaction and flag counts (by comment and by role), response length, and interviews coded using Braun & Clarke's reflexive thematic analysis [21]; two coders develop and refine themes collaboratively, with disagreements resolved through discussion.

E. Analysis. For **RQ1**, it is predicted that revisited responses will show higher IC than initial responses, tested with a linear mixed-effects model with participant and post as random effects; both an unadjusted model (total effect of the system) and a length-adjusted model (robustness check against IC gains reflecting longer rather than deeper responses) are reported, with Cohen's d alongside p -values. Since there are only six posts, a sensitivity analysis re-runs the model treating the stimulus posts as fixed effects. For **RQ2**, user-experience measures are summarized as averages, including how they trend across the six sessions — for instance, whether enjoyment or perceived effectiveness changes with repeated use. The end-of-study questionnaire, which asks fuller versions of the same questions, serves as a check on the lighter per-session ratings, alongside role-level usefulness ratings. For **RQ3**, patterns are examined across sessions: reaction and flag counts by agent role (e.g., whether "shifted my view" reactions become less common over time), response length, and themes from the interviews. Since agents always appear in the same order, differences between roles could reflect the role itself or just its position in the sequence. Role-level patterns are therefore reported as observations rather than as evidence that one role works better than another. After the study, two researchers check every flagged comment against outside sources and sort it as a genuine error, a defensible but debatable claim, or plain disagreement, then report how often the two agree.

F. Ethics and Open Science. All procedures will be reviewed by an institutional ethics board. Participants give informed consent and may withdraw at any time. Stimulus posts are presented with identifying details (names, handles, profile images) removed, except where the author is a public figure or institution speaking in that capacity. After the flag audit (Section 5E), participants receive a written debrief correcting any genuine factual errors in the comments they saw. The analysis plan will be pre-registered before data collection, and all materials and analysis code will be shared publicly.

6. Limitations

With one condition, the study cannot separate how much each part contributes — the cognitive roles, the communication style, or the act of revisiting. Without a control group that revisits without seeing agent comments, IC gains cannot be fully separated from just the usual effects of writing a second response — pausing, re-reading, or wanting to look thoughtful in a critical-thinking study. Since comments respond to each participant's initial view, content varies across people, and because agents appear in a fixed order, role differences are tangled with sequence position. Per-session ratings use one question each, keeping the six sessions manageable but making them less reliable. All participants come in wanting to improve their critical thinking; whether the effects hold for less motivated users is unknown. LLMs can invent facts or misrepresent views despite the safeguards in Section 3E; flagging catches errors users spot, but errors that slip past remain in the data. With ~24 participants, six researcher-selected posts, and one platform format, findings describe this system in this setting, not other users, posts, or interface styles. Finally, politically or emotionally charged and identity-related topics are excluded, leaving the hardest cases outside this study.

7. Expected Contribution

This work contributes (1) SoftVoicesAI, an interface where multiple AI agents help users reflect on social media takes. Each agent plays a distinct cognitive role that surfaces something the user might have overlooked — a hidden assumption, a missing perspective, or a leap in reasoning — and speaks in a communication style designed to help the reader stay open

and curious. (2) An empirical evaluation of whether using SoftVoicesAI is associated with increased integrative complexity, alongside findings on user experience and behavioral patterns. All materials, role and style specifications, and analysis code will be shared publicly.

REFERENCES

- [1] S. Milli et al., "Engagement, user satisfaction, and the amplification of divisive content on social media," *PNAS Nexus*, vol. 4, no. 3, pgaf062, 2025. <https://doi.org/10.1093/pnasnexus/pgaf062>
- [2] M. Cinelli, G. De Francisci Morales, A. Galeazzi, W. Quattrocioni, and M. Starnini, "The echo chamber effect on social media," *Proc. Natl. Acad. Sci. U.S.A.*, vol. 118, no. 9, e2023301118, 2021. <https://doi.org/10.1073/pnas.2023301118>
- [3] J. Haidt, "The emotional dog and its rational tail: A social intuitionist approach to moral judgment," *Psychol. Rev.*, vol. 108, no. 4, pp. 814–834, 2001. <https://doi.org/10.1037/0033-295X.108.4.814>
- [4] J. P. Dillard and L. Shen, "On the nature of reactance and its role in persuasive health communication," *Commun. Monogr.*, vol. 72, no. 2, pp. 144–168, 2005. <https://doi.org/10.1080/03637750500111815>
- [5] C. A. Bail et al., "Exposure to opposing views on social media can increase political polarization," *Proc. Natl. Acad. Sci. U.S.A.*, vol. 115, no. 37, pp. 9216–9221, 2018. <https://doi.org/10.1073/pnas.1804840115>
- [6] K. Yatani, Z. Sramek, and C.-L. Yang, "AI as Extraherics: Fostering higher-order thinking skills in human-AI interaction," arXiv:2409.09218, 2024. <https://doi.org/10.48550/arXiv.2409.09218>
- [7] V. Danry, P. Pataranutaporn, Y. Mao, and P. Maes, "Don't just tell me, ask me: AI systems that intelligently frame explanations as questions improve human logical discernment accuracy over causal AI explanations," in *Proc. CHI '23*, ACM, 2023. <https://doi.org/10.1145/3544548.3580672>
- [8] X. Fang, A. Xu, C.-L. Yang, Y.-F. Lin, S. Malacria, and K. Yatani, "LLM-based in-situ thought exchanges for critical paper reading," in *Proc. IUI '26*, ACM, 2026. <https://doi.org/10.1145/3742413.3789069>
- [9] Y. Liu, V. N. Shah, S. Suh, P. Siangliulue, T. August, and Y. Huang, "Perspectra: Choosing your experts enhances critical thinking in multi-agent research ideation," in *Proc. CHI '26*, ACM, 2026. <https://doi.org/10.1145/3772318.3791560>
- [10] T. Tanprasert, S. S. Fels, L. Sinnamon, and D. Yoon, "Debate chatbots to facilitate critical thinking on YouTube: Social identity and conversational style make a difference," in *Proc. CHI '24*, ACM, 2024. <https://doi.org/10.1145/3613904.3642513>
- [11] C.-W. Chiang, Z. Lu, Z. Li, and M. Yin, "Enhancing AI-assisted group decision making through LLM-powered devil's advocate," in *Proc. IUI '24*, ACM, 2024. <https://doi.org/10.1145/3640543.3645199>
- [12] D. Szabó et al., "Conversational inoculation to enhance resistance to misinformation," in *Proc. CHI '26*, ACM, 2026. <https://doi.org/10.1145/3772318.3790954>
- [13] T. H. Costello, G. Pennycook, and D. G. Rand, "Durably reducing conspiracy beliefs through dialogues with AI," *Science*, vol. 385, no. 6714, eadq1814, 2024. <https://doi.org/10.1126/science.adq1814>
- [14] T. Kriplean, J. Morgan, D. Freelon, A. Borning, and L. Bennett, "Supporting reflective public thought with ConsiderIt," in *Proc. CSCW '12*, ACM, 2012, pp. 265–274. <https://doi.org/10.1145/2145204.2145249>
- [15] C. G. Lord, M. R. Lepper, and E. Preston, "Considering the opposite: A corrective strategy for social judgment," *J. Pers. Soc. Psychol.*, vol. 47, no. 6, pp. 1231–1243, 1984. <https://doi.org/10.1037/0022-3514.47.6.1231>
- [16] B. Verheij, "Evaluating arguments based on Toulmin's scheme," *Argumentation*, vol. 19, no. 3, pp. 347–371, 2005. <https://doi.org/10.1007/s10503-005-4421-z>
- [17] R. M. Entman, "Framing: Toward clarification of a fractured paradigm," *J. Commun.*, vol. 43, no. 4, pp. 51–58, 1993. <https://doi.org/10.1111/j.1460-2466.1993.tb01304.x>
- [18] J. P. A. Ioannidis, "Why most published research findings are false," *PLoS Med.*, vol. 2, no. 8, e124, 2005. <https://doi.org/10.1371/journal.pmed.0020124>
- [19] G. Itzhakov, A. N. Kluger, and D. R. Castro, "I am aware of my inconsistencies but can tolerate them: The effect of high quality listening on speakers' attitude ambivalence," *Pers. Soc. Psychol. Bull.*, vol. 43, no. 1, pp. 105–120, 2017. <https://doi.org/10.1177/0146167216675339>
- [20] G. Baker-Brown et al., "The conceptual/integrative complexity scoring manual," in *Motivation and Personality: Handbook of Thematic Content Analysis*, C. P. Smith, Ed. Cambridge Univ. Press, 1992, pp. 401–418.
- [21] V. Braun and V. Clarke, "Using thematic analysis in psychology," *Qual. Res. Psychol.*, vol. 3, no. 2, pp. 77–101, 2006. <https://doi.org/10.1191/1478088706qp0630a>